Aviation Risk Analysis for Business Decisions

STEP1: Business Understanding

- Our company, Metro aviation is expanding.
- There is need to minimize operational and financial risk.
- The goal of this project is to identify aircraft types with the lowest safety risk based on historical accident data.

1.0 Investigation questions

- Which aircraft manufacturers have the lowest accident severity?
- Which aircraft categories show the lowest fatality rates?
- How has aviation safety changed over time?

STEP2: Data Understanding

- The dataset contains aviation accident records from 1962 to 2023. It includes aircraft manufacturer, model, accident date, injury counts, and damage severity. Due to the long time span and reporting differences, several fields may contain missing values.

STEP2: Data Understanding

- The dataset contains aviation accident records from 1962 to 2023.
- This includes aircraft manufacturer, model, accident date, injury counts, and damage severity.
- The aspects of data understanding include:

- 2.1 loading libraries
- 2.2 load data set
- 2.3 Inspect data
- 2.4 Check missing values

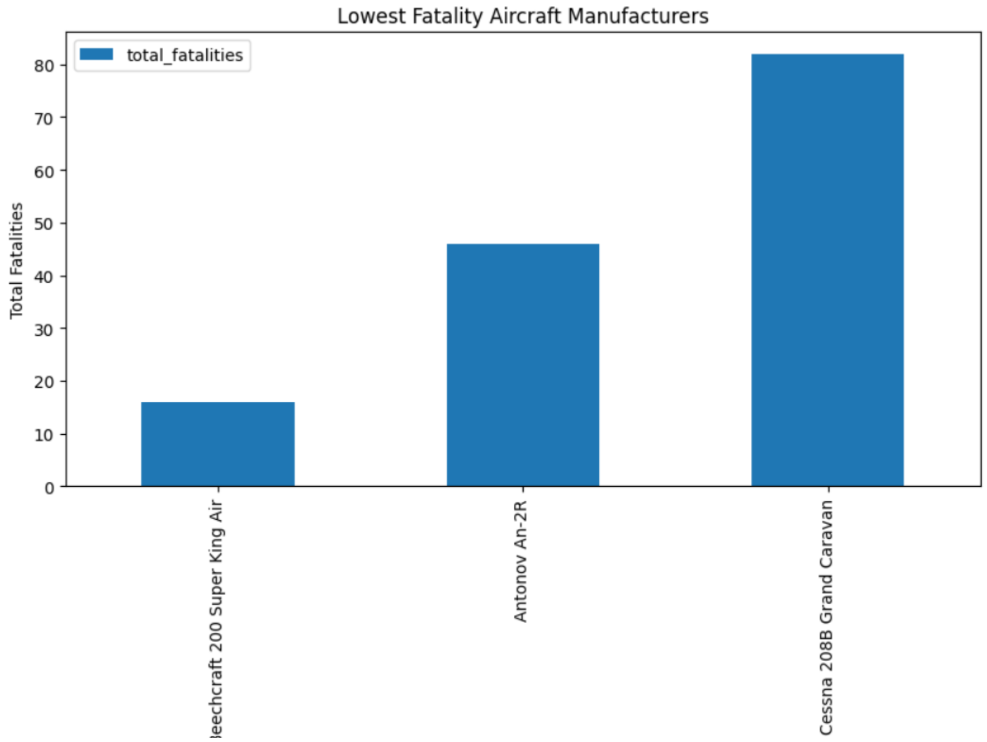
STEP3: Data Cleaning and Preparation (Pandas Focus)

- To prepare data for efficient analysis the following activities were done:

- 3.1 select relevant columns
- 3.2 Convert date
- 3.3 Handle missing data
- 3.4 Renaming aviation columns

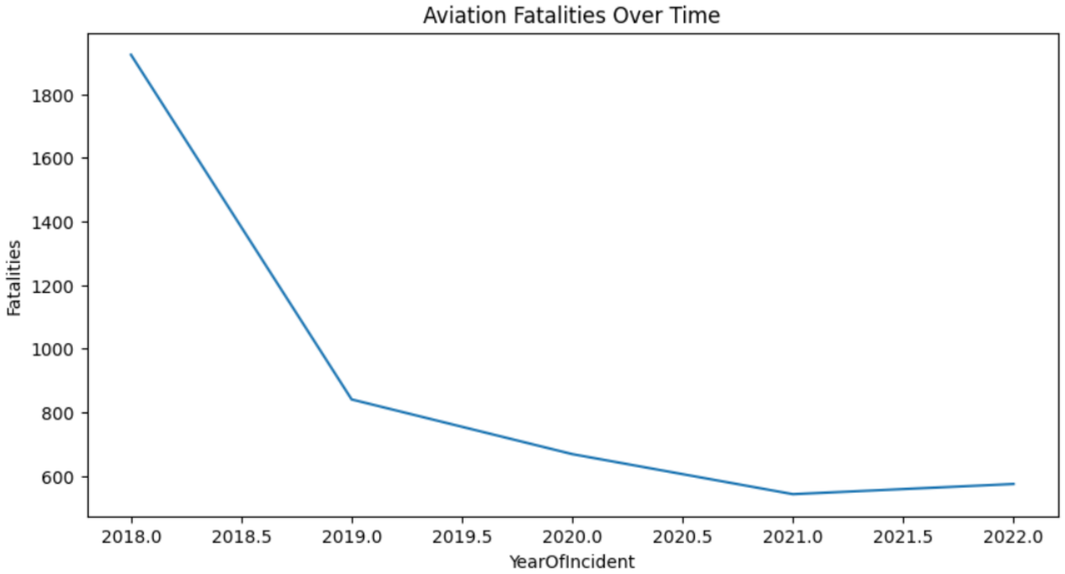
STEP4: Data Analysis (Answer business questions)

4.1 Which manufacturers have the lowest fatality risk?



- Beechcraft 200 Super King Air reports the least fatalities at below 20 and is therefore the safest.
- Other models reporting low fatalities are Antonov An-2R and Cessna 208B Grand Caravan.
- This analysis recommends the three for safety passenger safety is this is to be the top consideration.

4.3 How has aviation safety changed over time?



- We notice that overally, fatalities have reduced.
- This can be attributed to improved technologies and efficiently managed flight safety.
- So irrespective of the model, there is a down ward trend in times of injuries and deaths to passengers